

YINGJIN LI

M.Phil. Student, The Hong Kong University of Science and Technology (Guangzhou)

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Research Interests: Visual Anomaly Detection · Open-World Visual Understanding
Multimodal Learning · Vision-Language Models

Education

The Hong Kong University of Science and Technology (Guangzhou)	
M.Phil., Data Science and Analytics	Sep 2025 - Jun 2027 (expected)
Supervisor: Prof. Lei Zhu	
Recipient of the Red Bird Scholarship (RMB 240,000).	
Wuhan Textile University	
B.Eng., Data Science and Big Data Technology	Sep 2021 - Jun 2025
Average score: 86.7/100 · Academic ranking: Top 5% · Overall ranking: 1st in major	

Publication List

1. ROAD: Rule-Grounded Context-Aware Open-World Driver Anomaly Detection	
Yingjin Li, Shuaibo Li, Wei Ma, Zhijie Qiu, Fuxiang Zhai, Sixiang Chen, Zhaolu Kang, and Lei Zhu. <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2026 · Accepted	
2. Fashion Image Retrieval Based on Multimodal Features Enhancement and Fusion	
Yingjin Li, Shufan He, Zhaojing Wang, Jin Huang, Xinrong Hu, and Li Li. <i>Computer Animation and Social Agents (CASA 2024)</i> · Published · Springer, CCIS 2374, pp. 118-131, 2025.	
3. PriSM-AD: Learning Common Normal Priors and Anomaly Semantics for Diffusion-Based Anomaly Detection	
Yingjin Li, Sixiang Chen, Wenxue Li, Shuaibo Li, Fugee Tsung, and Lei Zhu. <i>IEEE Transactions on Multimedia (TMM)</i> · Under Review	

Research Experience

ROAD · Multimodal Driver Anomaly Detection [1]	Apr 2026 - Present
- Designed and implemented a multimodal anomaly detection framework that integrates driver video, driving context, and explicit behavioral rules to identify previously unseen driver anomalies. - Developed rule-grounded reasoning that translates domain knowledge into structured guidance for multimodal large language models, supporting explanations of why a behavior is abnormal beyond predefined action categories. - Integrated context-aware visual reasoning to distinguish visually similar behaviors across driving situations and account for context-dependent safety rules in open-world anomaly understanding. - Built an evaluation pipeline across driver-monitoring datasets with free-form anomaly descriptions, human assessments, and MLLM baselines to assess open-world generalization and model-human agreement.	
PriSM-AD · Diffusion-Based Anomaly Detection [3]	
- Designed a two-stage diffusion framework that separates learning common normal visual priors from anomaly localization, combining normality modeling with semantic cues for industrial defect detection. - Developed prior-guided reconstruction to recover normal appearance while suppressing anomalous structures, providing a normal reference for identifying subtle defects. - Integrated anomaly semantics and reconstruction cues to support both image-level anomaly detection and pixel-level localization across industrial categories. - Analyzed reconstruction quality, cross-category generalization, anomaly localization, and inference efficiency.	

Honors & Awards

- National Scholarship, Ministry of Education of China, 2024
- First Prize, Hubei Province, China Mathematical Contest in Modeling
- National First Prize, Teddy Cup Data Mining Challenge
- National First Prize, Huazhong Cup Mathematical Modeling Challenge